THULASIRAM G

ROS2 Robotics Engineer | Deployment & Simulation | Nav2 · SLAM · Gazebo · Autonomous Navigation

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PROFESSIONAL SUMMARY

Final-year Robotics & Automation Engineer (CGPA 8.8, Class Topper) with hands-on ROS2 Humble experience across full autonomous-stack development — SLAM mapping, Nav2 path planning, Gazebo simulation, and real LiDAR/ultrasonic sensor integration. Shipped a complete ROS2 café-delivery robot (SLAM + Nav2 + order-management state machine) end-to-end, from URDF design through debugging navigation, threading, and controller-tuning issues to a working autonomous demo. Backed by 3 industrial internships spanning ROS2 robotics, PLC-driven manufacturing automation, and live production-floor robotics across 12+ stations. Targeting Deployment Engineer / ROS2 Robotics Engineer roles building and shipping autonomous systems in robotics manufacturing environments.

EDUCATION

B.E. Robotics & Automation Engineering — Dhaanish Ahmed College of Engineering
CGPA: 8.8/10 (Class Topper) | 2022 – 2026

Higher Secondary Certificate, Computer Science — KSR Matric Hr. Sec. School
88.8% | 2020 – 2022

CORE COMPETENCIES

- **Robotics & Simulation:** ROS2 Humble, Nav2 (path planning, costmaps, controller tuning), SLAM (slam_toolbox), Gazebo, RViz2, URDF/SDF modelling, LiDAR & ultrasonic sensor integration, autonomous navigation, obstacle avoidance
- **Software Engineering:** Python (rclpy), HTML, CSS, multithreading & concurrency (threading.Lock), colcon build system, state-machine design, debugging & root-cause analysis, Git/GitHub
- **Automation & Controls:** PLC ladder logic, DELTA PLC software, industrial I/O mapping, field wiring, embedded systems (Embedded C, Arduino, RFID, relay control, debounce logic)
- **Tools:** ros2 CLI, rqt_graph, ros2 topic echo, ros2 service call, colcon build, Fusion 360, AutoCAD

INTERNSHIP EXPERIENCE

ROS2 Robotics Intern | *Evolve Robot Lab* 2026 | Chennai

- Built a full-stack ROS2 Humble delivery robot in Gazebo integrating SLAM mapping, Nav2 navigation, and a custom order-management state machine, taking it from initial design through to a working end-to-end simulation.
- Fixed a critical order-tracking bug where shared state was overwritten on concurrent orders, and resolved a threading race condition in the control node using threading.Lock().
- Resolved navigation-goal and AMCL localisation failures by re-tuning costmap inflation zones against the map geometry and eliminating a faulty /initialpose publisher loop.
- Tuned RegulatedPurePursuitController parameters, fixing a YAML configuration error and raising desired_linear_vel to increase navigation speed while maintaining path-following stability.
- Completed structured ROS2 Humble training on nodes, topics, services, and action servers, with live sensor and robot-state visualisation in RViz2.

PLC Automation Intern | *I/O LOGIX* 2024 | Chennai

- Programmed and commissioned industrial ladder-logic control panels using DELTA PLC software for real-world automation control projects.
- Executed field wiring and I/O mapping for automation panels, gaining direct exposure to industrial control-system architecture end-to-end.

Automation Intern | *Kosei Minda Aluminum Company* 2023 | Oragadam

- Studied smart-manufacturing workflows across 12+ robotic stations in a live alloy-wheel production plant, analysing integration between industrial robotic arms and PLC automation.
- Gained applied exposure to lean-manufacturing and automated quality-control principles in a high-volume production environment.

KEY PROJECTS

Butler Robot — Autonomous Café Delivery Robot

- Built an autonomous café-delivery robot in ROS2 Humble combining SLAM mapping, Nav2 path planning, and a custom order-management state machine to deliver orders to tables without manual intervention, taking the system from a fresh workspace to a working end-to-end simulation in Gazebo.
- Generated a SLAM occupancy map of the café environment using slam_toolbox and validated it against a handcrafted reference map for navigation-grade accuracy, then tuned costmap inflation zones to fix repeated navigation-goal placement failures near café furniture.
- Diagnosed and fixed critical bugs — an order-ID overwrite causing wrong-order confirmations, a threading race condition resolved with threading.Lock(), and a looping /initialpose publisher that was resetting AMCL localisation — stabilising the robot for a repeatable delivery demo.

Tech Stack: *ROS2 Humble, Nav2, slam_toolbox, Gazebo, RViz2, Python, rclpy, threading*

Obstacle Avoidance Robot

- Built a fully autonomous obstacle-avoidance robot in ROS2, subscribing to /scan (LaserScan) and publishing threshold-based reactive velocity commands to /cmd_vel in real time to detect and steer around obstacles.
- Simulated the full robot environment in Gazebo and validated live sensor data and robot state in RViz2 before behaviour tuning.

Tech Stack: *ROS2 Humble, Python, rclpy, Gazebo, RViz2, LiDAR (LaserScan)*

LiDAR Car Simulation in Gazebo

- Simulated a car model in Gazebo with an integrated LiDAR sensor using ROS2 middleware for autonomous-navigation research, configuring URDF/SDF robot description files to publish real-time laser-scan data over ROS2 topics.
- Visualised LiDAR point-cloud data in RViz2 to validate environment perception for downstream navigation algorithms.

Tech Stack: *ROS2 Humble, Python, Gazebo, URDF, RViz2*

Disaster Response Robot Simulation — Computer Vision-Driven (Software)

- Developed a software simulation of an AI-driven disaster-response robot that uses computer vision to detect victims and hazards in a simulated environment, feeding detections into the robot's decision logic to prioritise areas for response.
- Built the perception pipeline to process simulated camera input and flag detected victims/hazards in real time, forming the basis for downstream path-planning and response prioritisation.

Tech Stack: *Python, Computer Vision (OpenCV/CNN-based detection), simulation environment*

Agricultural UAV — Autonomous Pesticide Spraying

- Designed a GPS-guided UAV with relay-driven spray-nozzle actuation and automated mission logic to improve field spray-coverage accuracy and reduce farmer chemical exposure.

Embedded RFID Access & Motor Control Systems

- Delivered two embedded-control systems in Arduino/Embedded C: an RFID door-access controller with relay-driven locking, and a debounce-protected RFID motor start/stop controller.

Tech Stack: *Embedded C, Arduino, RFID Module, Relay*

LANGUAGES

Tamil (Native), English (Professional), Telugu (Conversational)